

Technology Stack

Date	4 July 2026
Team ID	xxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3, Bootstrap	Used to develop a responsive and user-friendly web interface for entering agricultural parameters and displaying crop recommendations. Bootstrap ensures a mobile-friendly and consistent UI.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python 3.10+, Flask	Flask provides a lightweight framework for integrating the machine learning model with the web application. Python supports efficient data processing and machine learning implementation.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset, Pickle (.pkl)	The agricultural dataset is stored in CSV format, while the trained machine learning model is saved as a Pickle (.pkl) file for fast prediction without retraining.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Flask Development Server (Local Deployment)	The application is deployed locally using the Flask development server for testing, demonstration, and real-time crop prediction. The architecture can be extended to cloud platforms in the future.

5	Version Control & CI/CD	Git, GitHub	Git is used for version control, and GitHub enables source code management, collaboration among team members, and project tracking.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, PyCharm, Anaconda Navigator	Scikit-learn is used for machine learning model development, Pandas and NumPy for data preprocessing, Matplotlib and Seaborn for data visualization, Jupyter Notebook for experimentation, and PyCharm/Anaconda for project development and dependency management.